Reads over Atom Spaces: A Unified Mathematical Formulation of Context, Retrieval, and Parameters in Language Models

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Abstract

Finite attention, external retrieval, and parameter-expert routing share a common operational core: computing a query-conditioned convex combination of value vectors in an output representation space. We formalize this mechanism through the analytical lens of a valued atom space $(\Omega, \mathcal{F}, \nu, \upsilon)$ and a Bochner read operator $R(x; \Omega) = \int_\Omega \upsilon(\omega)\, d\mu_x(\omega)$. While continuous integration provides a general mathematical umbrella, practical neural architectures specialize this Bochner integral to discrete counting measures, yielding softmax-weighted reads across token context, external corpora, and parameter experts. Under this framework, we analyze: (i) a Lipschitz-margin condition sufficient for newly inserted atoms to enter an exact top-k set, which we empirically validate on our benchmark; (ii) exact ranked top-k truncation bounds showing exponential tail decay; and (iii) signed reference measures where anti-atoms algebraically cancel external-read contributions. Controlled synthetic and CounterFact studies confirm that while the formulation provides certified error budgets and exact cancellation under perfect alignment, practical retrieval faces real trade-offs: exact tail certification requires computing all scores, approximate signed cancellation degrades rapidly under key mismatch, and uncalibrated activation thresholds discard valid paraphrases. Replacing static thresholds with data-driven ROC calibration ($\tau^* = 0.55$, calibration AUC = 0.817) increases activated paraphrase recall from 38.2% to 53.2% (with neighborhood false activation at 16.1%) on CounterFact-500. Across five seeds, surface-matched routing accuracies are 100.0 ± 0.0%, 82.4 ± 0.9%, and 100.0 ± 0.0%. Matched negative atoms cancel paired signed-read contri-

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butions (2.0407 ± 0.00003% residual probability on a 49-token vocabulary). A ten-fact GPT-2 mechanism study evaluates factual steering alongside naive fine-tuning and rank-one (ROME-lite) baselines, complemented by an expanded evaluation across $N = 50$ CounterFact records under calibrated thresholding ($\tau^* = 0.55$), achieving 100.0% rewrite efficacy and raising paraphrase score from 22.0% [12.0%, 34.0%] to 66.0% [52.0%, 78.0%] (56.0% exact argmax). We position the framework against retrieval-augmented generation and memory-based editors (SERAC, GRACE).

Keywords: Language models, Atom spaces, Measure theory, External memory, Vector retrieval, Expert routing, Model editing

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1. Introduction

Autoregressive language models encode linguistic abstractions, syntax, and factual knowledge within dense parameter matrices [1, 2, 3]. While gradient-based optimization successfully acquires structural linguistic generalizations, compiling dynamic factual statements directly into static weights presents persistent operational challenges. Real-world facts obsolesce rapidly, require targeted localized updates, and are subject to regulatory erasure mandates (e.g., the "Right to be Forgotten"). Consequently, post-hoc weight modification often triggers catastrophic interference, parameter drift, or high computational costs [4, 5].

To address the rigidity of dense parameter storage, multiple architectural strategies have emerged across distinct sub-disciplines:

- *In-context reasoning* caches dynamic token representations in the key-value sequence memory [1].
- *Retrieval-augmented generation* (RAG) and nearest-neighbor language models (kNN-LM) decouple memory by querying external vector databases of textual passages [6, 7, 8].
- *Modular computation* mechanisms, including Mixture-of-Experts (MoE) and parameter-efficient adapter banks, dynamically route tokens across banks of specialized sub-networks [9, 10, 11].

While these techniques all rely on key-value similarity matching [3, 12, 13], they are typically formulated using disconnected notations, disparate routing heuristics, and distinct operational semantics.

In this work, we present a unified mathematical formulation for in-context attention, external retrieval, and parameter-expert output routing. By expressing each mechanism through an integral of a query-conditioned probability measure against a value map over a valued atom space, we establish a shared analytical umbrella. In discrete neural architectures, this Bochner read operator specializes algebraically to query-conditioned softmax-weighted convex combinations over value representations.

Through this lens, we make the following contributions:

1. **A Unified Mathematical Formulation:** We define the read operator $R(x; \Omega)$ over valued atom spaces $(\Omega, \mathcal{F}, \nu, v)$ and formulate an integrated tri-space architecture that routes between context (Ω_{ctx}), corpus retrieval (Ω_{corpus}), and parameter experts (Ω_{params}).

2. **A Checkable Retrieval Condition with Empirical Validation:** We derive a Lipschitz-margin condition sufficient for a novel atom to enter an exact top-k set and validate it empirically on a controlled benchmark using analytical and empirical Lipschitz estimates.

3. **Softmax Tail Truncation Bounds:** We prove Theorem 1, establishing two exponential tail bounds on exact ranked top-k reads, and formulate Corollary 1 as an error-budgeted stopping rule.

4. **Signed External-Memory Suppression:** By extending atom spaces to signed reference measures ($\nu = \nu^+ - \nu^-$), we formalize exact zero-retraining cancellation of external-memory contributions via anti-atoms, accompanied by an approximate cancellation perturbation bound.

5. **Controlled Benchmark Evaluations:** Across five seeds, we evaluate lifelong cumulative insertion, multi-space routing, approximate cancellation stress tests, and CounterFact retrieval with HNSW candidate indexing and ROC threshold calibration. We compare against naive fine-tuning, ROME-lite, covariance-regularized CounterFact-scale ROME, and memory-based editing paradigms (SERAC, GRACE).

2. From a Discrete Function to an Integral

2.1. The Token-Level Function

A language model is conventionally defined as a conditional probability distribution over a finite vocabulary V of size d, given an input sequence. Fixing a maximum context capacity c and introducing a null padding symbol $\varnothing$ yields an augmented vocabulary $\bar{V} = V \cup \{\varnothing\}$. An input sequence is represented as a tuple

$$x = (x_1, \ldots, x_c) \in \bar{V}^c,$$

where unassigned sequence positions are padded with $\varnothing$. A language model is a mapping

$$p : \bar{V}^c \to \Delta^{d-1}, \qquad \Delta^{d-1} = \left\{ z \in \mathbb{R}^d : z_i \geq 0, \sum_{i=1}^{d} z_i = 1 \right\}.$$

2.2. Attention as an Integral

Let $\phi : \bar{V} \to \mathbb{R}^e$ denote an embedding map with $\phi(\varnothing) = 0$. Extending the sequence to a step function over the continuous interval $[0, c]$ gives:

$$f_x(t) = \phi\big(x_{\lceil t \rceil}\big), \qquad t \in (0, c],$$

with $f_x(0) = \phi(x_1)$. Let $\alpha_x : [0, c] \to \mathbb{R}_{\geq 0}$ be an attention weight density satisfying $\int_0^c \alpha_x(t)\, dt = 1$, with $\alpha_x(t) = 0$ whenever $x_{\lceil t \rceil} = \varnothing$. The context representation vector is:

$$z(x) = \int_0^c \alpha_x(t)\, f_x(t)\, dt \in \mathbb{R}^e,$$

and the corresponding token prediction distribution is $p(x) = \mathrm{softmax}\big(W z(x) + b\big)$. Because f_x is piecewise constant over unit intervals, the continuous integral evaluates algebraically to the standard discrete attention weighted sum [1, 14]:

$$\int_0^c \alpha_x(t)\, f_x(t)\, dt = \sum_{i=1}^{c} \left(\int_{i-1}^{i} \alpha_x(t)\, dt \right) \phi(x_i) = \sum_{i=1}^{c} \alpha_i \phi(x_i).$$

This continuous formulation naturally generalizes by replacing the sequence index interval $[0, c]$ with an arbitrary measurable *atom space*.

3. Atom Spaces: A Common Object for Context, Retrieval, and Parameters

Definition 1 (Valued Atom Space)**.** *A* valued atom space *is a tuple* $(\Omega, \mathcal{F}, \nu, v)$*, where* $(\Omega, \mathcal{F})$ *is a measurable space of elements designated as* atoms*,* ν *is a* σ*-finite reference measure, and* $v : \Omega \to \mathbb{R}^e$ *is a measurable* value map *into a common output representation space.*

Table 1: Three structural realizations of the unified read operator $R(x; \Omega)$.

Instance	**Atom Space** Ω	**Atom** ω	**Value Map** $v(\omega)$	**Operational Equivalent**
Context	$\{1, \ldots, c\}$	Token index	Embedding $\phi(x_\omega)$	In-context self-attention [1]
Corpus	$\mathcal{D}$ (mutable)	Factual record	Encoded fact vector	Dense retrieval / kNN-LM [7]
Parameters	Θ (expert bank)	Modular expert	Expert output $g_\omega(x) \in \mathbb{R}^e$	MoE-style output routing [9, 10]

Definition 2 (Query-Conditioned Kernel and Read Operator)**.** *Given a query $x \in \mathcal{X}$, let $K_x : \Omega \to \mathbb{R}_{\geq 0}$ be measurable with*

$$0 < Z_x := \int_\Omega K_x(\omega)\, d\nu(\omega) < \infty.$$

Then K_x induces a query-conditioned probability measure μ_x on $(\Omega, \mathcal{F})$ via the Radon–Nikodym derivative:

$$d\mu_x(\omega) = \frac{K_x(\omega)}{Z_x}\, d\nu(\omega).$$

If $\int_\Omega \|v(\omega)\|_2\, d\mu_x(\omega) < \infty$, the read *conditioned on x is the Bochner integral:*

$$R(x; \Omega) = \int_\Omega v(\omega)\, d\mu_x(\omega) \in \mathbb{R}^e.$$

When ν is a discrete counting measure over a finite set $\Omega = \{\omega_1, \ldots, \omega_N\}$, the Bochner integral evaluates directly to a softmax-weighted sum:

$$R(x; \Omega) = \sum_{i=1}^{N} \frac{K_x(\omega_i)}{\sum_{j=1}^{N} K_x(\omega_j)} v(\omega_i).$$

As summarized in Table 1, in-context attention, dense retrieval, and expert-output routing are structural realizations of this read operator, differing in their domain of integration and the query-dependence of their values.

3.1. The Unified Architecture

Context, external corpus memory, and parameter routing combine into an integrated representation:

$$z(x) = \lambda_{\text{ctx}}(x) R(x; \Omega_{\text{ctx}}) + \lambda_{\text{corpus}}(x) R(x; \Omega_{\text{corpus}}) + \lambda_{\text{params}}(x) R(x; \Omega_{\text{params}}),$$

$$p(x) = \mathrm{softmax}\big(W_0\, z(x) + b_0\big),$$

where $\lambda(x) \in \Delta^2$ is an input-dependent gating distribution across the three spaces, and $W_0 \in \mathbb{R}^{d\times e}$, $b_0 \in \mathbb{R}^d$ represent the linguistic projection head. For parameter experts, the value is query-dependent, $v_x(\omega) = g_\omega(x)$; the read is therefore an output-space mixture, not an average of parameter tensors.

3.2. System, Calibration, and Tying Principles

- **Scale Mismatch and Norm Calibration:** The cardinalities of the spaces vary across orders of magnitude ($|\Omega_{\text{ctx}}| \sim 10^3$, $|\Omega_{\text{params}}| \sim 10^1$, $|\Omega_{\text{corpus}}| \sim 10^9$). Normalizing a single joint softmax across the disjoint union $\Omega_{\text{ctx}} \sqcup \Omega_{\text{corpus}} \sqcup \Omega_{\text{params}}$ causes the largest space to artificially dominate probability mass. Separate intra-space measure normalization combined with calibrated convex gating $\lambda(x)$ is required. Crucially, key representations across distinct spaces must share identical geometric scales (unit L_2-normalization) to prevent dot-product magnitudes from skewing the router toward unnormalized representations.

- **Tied Value-to-Token Embeddings:** In token generation, the read vector $z_{\text{corpus}}(x)$ should project onto target vocabulary tokens. Letting $E \in \mathbb{R}^{d\times e}$ denote the vocabulary embedding matrix, we tie values via $v(\omega) = E[y_\omega]/\|E[y_\omega]\|$ and evaluate logits as Ez/τ. Isolating atom ω^* aligns $z(x)$ directly with the target row.

3.3. Signed Atom Spaces and the Hahn–Jordan Decomposition

To support zero-retraining cancellation in external memory, we extend the reference measure ν to a *signed measure*. By the Hahn–Jordan Decomposition Theorem, any signed reference measure uniquely decomposes into mutually singular non-negative measures:

$$\nu = \nu^+ - \nu^-, \qquad \nu^+ \perp \nu^-,$$

where ν^+ governs positive constructive atoms and ν^- governs negative "anti-atoms". Defining the total variation measure $|\nu| = \nu^+ + \nu^-$, we assume $0 < Z_x^\pm = \int_\Omega K_x \, d|\nu| < \infty$. The normalized signed query measure is:

$$d\mu_x^\pm(\omega) = \frac{K_x(\omega)}{Z_x^\pm}\big(d\nu^+(\omega) - d\nu^-(\omega)\big).$$

Total variation is normalized to unity, while the signed integral total mass varies in $[-1, 1]$.

4. Mathematical Conditions for Post-Hoc Knowledge Editing

Let the query-key kernel be a scaled inner product:

$$K_x(\omega) = \exp\left(\frac{\langle \phi_q(x), \phi_k(\omega)\rangle}{\tau}\right),$$

where $\phi_q : \mathcal{X} \to \mathbb{R}^m$ and $\phi_k : \Omega \to \mathbb{R}^m$ are query and key encoders, and $\tau > 0$ is a temperature parameter. Base training optimizes ϕ_q and ϕ_k on an initial pool $\Omega_{\text{train}} \subset \Omega$. Post-hoc editing requires that inserting a novel atom $\omega^* \notin \Omega_{\text{train}}$ at test time reliably captures probability mass $\mu_x(\omega^*)$ for associated queries x^*, without modifying encoder parameters.

4.1. Kernel Locality

Definition 3 (Metric Locality)**.** *The key encoder ϕ_k is L-Lipschitz continuous with respect to a semantic pseudo-metric d_Ω on Ω if*

$$\|\phi_k(\omega) - \phi_k(\omega')\|_2 \le L\, d_\Omega(\omega, \omega') \qquad \forall\, \omega, \omega' \in \Omega.$$

Proposition 1 (Score Stability on Unseen Atoms)**.** *Let ϕ_k be L-Lipschitz with respect to d_Ω. For any training anchor $\omega_0 \in \Omega_{\text{train}}$ and any newly inserted atom $\omega^* \in \Omega$:*

$$\left|\langle \phi_q(x), \phi_k(\omega^*)\rangle - \langle \phi_q(x), \phi_k(\omega_0)\rangle\right| \;\le\; \|\phi_q(x)\|_2\, L\, d_\Omega(\omega^*, \omega_0).$$

Proof. By the Cauchy–Schwarz inequality and the Lipschitz condition:

$$\begin{aligned}\left|\langle \phi_q(x), \phi_k(\omega^*) - \phi_k(\omega_0)\rangle\right| &\le \|\phi_q(x)\|_2\, \|\phi_k(\omega^*) - \phi_k(\omega_0)\|_2 \\ &\le \|\phi_q(x)\|_2\, L\, d_\Omega(\omega^*, \omega_0).\end{aligned}$$

□

4.2. The Retrieval Margin Condition

Let $\tau_k(x) = \max^{(k)}_{\omega\in\Omega_{\text{train}}} \langle \phi_q(x), \phi_k(\omega)\rangle$ denote the k-th highest score among existing training atoms. For the novel atom ω^* to enter the top-k retrieved set under query x, it must satisfy $\langle \phi_q(x), \phi_k(\omega^*)\rangle > \tau_k(x)$. Combining this with Proposition 1 yields a checkable sufficient condition:

$$\langle \phi_q(x), \phi_k(\omega_0)\rangle - \|\phi_q(x)\|_2\, L\, d_\Omega(\omega^*, \omega_0) \;>\; \tau_k(x), \tag{1}$$

where ω_0 is a training anchor semantically proximal to ω^*. Equation (1) is a sufficient condition for top-k inclusion; successful prediction additionally depends on retrieved probability mass, value representation, output margin, and specificity on controls.

Empirical Verification and Conservatism of the Margin Condition. We empirically evaluate Equation (1) on our benchmark. We instantiate $d_\Omega(\omega, \omega') = \|\phi_0(\omega) - \phi_0(\omega')\|_2$ using the Euclidean distance between frozen MiniLM backbone embeddings. For the residual adapter $\phi(x) = \mathrm{Normalize}(x + 0.1 \cdot \mathrm{MLP}(x))$, the operator norm product yields an analytical upper bound $L \leq 2.105$, while the empirical maximum directional ratio across the corpus is $\hat{L} = 2.236$. Evaluating Equation (1) across all 40 insertion facts using their nearest training anchors ω_0 reveals a stark contrast between empirical reality and worst-case certification: while actual top-1 entry is 100.0%, the certified lower bound guarantees entry for 0.0% of records across top-1, top-5, and top-10. This occurs because Cauchy–Schwarz assumes worst-case adversarial alignment in 384 dimensions; with inter-fact semantic distances averaging $d_\Omega \approx 0.78$, the worst-case penalty $L \cdot d_\Omega \approx 1.74$ easily overwhelms the inner-product score margin (LHS ≈ -1.34 vs. $\tau_k \approx 0.35$). This empirical test demonstrates that while Equation (1) is mathematically sound, global Lipschitz certificates are overly conservative in sparse semantic corpora, proving that local or directional Lipschitz bounds are necessary for non-vacuous retrieval guarantees.

4.3. Anti-Atoms and Exact External-Read Cancellation

Proposition 2 (Matched-Pair Cancellation)**.** *Let an atom $\omega^* \in \Omega_{\text{corpus}}$ encode a target fact with tied token value $v^* = E[y^*]/\|E[y^*]\|$. If an anti-atom ω^- is inserted into the signed atom space with identical key $\phi_k(\omega^-) = \phi_k(\omega^*)$, identical value $v^- = v^*$, and equal positive masses $m^* := \nu^+(\{\omega^*\}) = m^- := \nu^-(\{\omega^-\}) >$* 0*, then for any query x:*

$$R^\pm(x; \{\omega^*, \omega^-\}) = \mathbf{0}.$$

Proof. Under the signed measure integral of Section 3.3:

$$\int_{\{\omega^*, \omega^-\}} v(\omega)\, d\mu_x^\pm(\omega) = \frac{K_x(\omega^*) m^* v^* - K_x(\omega^-) m^- v^*}{Z_x^\pm} = \frac{K_x(\omega^*) m^* (v^* - v^*)}{Z_x^\pm} = \mathbf{0}.$$

This establishes exact cancellation of the matched pair's contribution to the signed external read. □

Proposition 2 does not imply that the complete model representation is zero: background atoms, context, parameter experts, residual model states, and output bias remain. The positive record and anti-atom remain stored, and the proposition does not claim removal of knowledge compiled into dense pretrained parameters.

Proposition 3 (Approximate Pair Cancellation)**.** *Let $a = K_x(\omega^*)\nu^+(\{\omega^*\})$, $b = K_x(\omega^-)\nu^-(\{\omega^-\})$, and let $Z_x^\pm > 0$ be the total-variation normalization. For positive value v and anti-atom value $\tilde{v}$, the residual pair contribution satisfies:*

$$\left\| \frac{av - b\tilde{v}}{Z_x^\pm} \right\|_2 \leq \frac{|a-b|\,\|v\|_2 + b\,\|v - \tilde{v}\|_2}{Z_x^\pm}.$$

Proof. Write $av - b\tilde{v} = (a-b)v + b(v - \tilde{v})$ and apply the triangle inequality, then divide by $Z_x^\pm$. □

Exact cancellation is sensitive to key-induced kernel mismatch, unequal mass, and value mismatch. Paraphrases affect cancellation only through their effect on kernel weights; identical keys cancel for all queries.

5. Computational Tractability and the Softmax Tail Truncation Bound

Evaluating $R(x;\Omega) = \int_\Omega v(\omega)\, d\mu_x(\omega)$ densely over a large corpus is computationally expensive. We analyze numerical truncation to the exact ranked top-k elements:

$$\hat{R}_k(x;\Omega) = \sum_{i=1}^{k} v(\omega_{(i)})\, \frac{K_x(\omega_{(i)})}{\sum_{j=1}^{k} K_x(\omega_{(j)})}.$$

In this discrete setting, numerical truncation corresponds to numerical quadrature of the query-conditioned measure against the value map, omitting the low-probability tail $Z_{\text{tail}} = \sum_{i=k+1}^{N} e^{s_{(i)}/\tau}$. Dense scoring over N items costs $\mathcal{O}(Ne)$ before the top-k read. When paired with an ANN index, aggregation costs:

$$\mathcal{O}(c \cdot e) + \mathcal{O}(k_{\text{corpus}} \cdot e) + \mathcal{O}(k_{\text{params}} \cdot e).$$

Let $N = |\Omega|$, and $V_{\max} := \sup_{\omega\in\Omega} \|v(\omega)\|_2 < \infty$. For query x, let $s_i := \langle \phi_q(x), \phi_k(\omega_i)\rangle$, with ordering $s_{(1)} \geq \cdots \geq s_{(N)}$. Define $Z := \sum_{j=1}^{N} e^{s_{(j)}/\tau}$ and $Z_k := \sum_{j=1}^{k} e^{s_{(j)}/\tau}$. Define the retrieval spectral gap at rank k as $\Delta_k(x) := s_{(1)} - s_{(k+1)} \geq 0$.

Theorem 1 (Softmax Tail Truncation Error Bound)**.** *For any query x and truncation depth $k \in \{1, \ldots, N-1\}$, the Euclidean approximation error of top-k truncation satisfies:*

$$\|R(x;\Omega) - \hat{R}_k(x;\Omega)\|_2 \;\leq\; 2V_{\max} \cdot (N-k) \cdot \exp\left(-\frac{\Delta_k(x)}{\tau}\right).$$

An alternative bound is:

$$\|R(x;\Omega) - \hat{R}_k(x;\Omega)\|_2 \;\le\; 2V_{\max} \cdot \left(\frac{N-k}{k}\right) \cdot \exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right).$$

Proof. Decompose the difference between the full read and the top-k truncation:

$$\begin{aligned} R(x;\Omega) - \hat{R}_k(x;\Omega) &= \sum_{i=1}^{k} v_{(i)} e^{s_{(i)}/\tau}\left(\frac{1}{Z} - \frac{1}{Z_k}\right) + \sum_{i=k+1}^{N} v_{(i)} \frac{e^{s_{(i)}/\tau}}{Z} \\ &= \frac{Z_k - Z}{Z Z_k} \sum_{i=1}^{k} v_{(i)} e^{s_{(i)}/\tau} + \frac{1}{Z} \sum_{i=k+1}^{N} v_{(i)} e^{s_{(i)}/\tau}. \end{aligned}$$

Let $Z_{\text{tail}} := Z - Z_k = \sum_{i=k+1}^{N} e^{s_{(i)}/\tau}$. Notice $\frac{1}{Z_k}\sum_{i=1}^{k} v_{(i)} e^{s_{(i)}/\tau} = \hat{R}_k(x;\Omega)$, and define $R_{\text{tail}} := \frac{1}{Z_{\text{tail}}}\sum_{i=k+1}^{N} v_{(i)} e^{s_{(i)}/\tau}$. Substituting yields:

$$R(x;\Omega) - \hat{R}_k(x;\Omega) = \frac{Z_{\text{tail}}}{Z}\big(R_{\text{tail}} - \hat{R}_k(x;\Omega)\big).$$

Taking norms and applying the triangle inequality:

$$\|R(x;\Omega) - \hat{R}_k(x;\Omega)\|_2 \le \frac{Z_{\text{tail}}}{Z}\big(\|R_{\text{tail}}\|_2 + \|\hat{R}_k(x;\Omega)\|_2\big) \le 2V_{\max}\frac{Z_{\text{tail}}}{Z}.$$

Since $Z \ge e^{s_{(1)}/\tau}$ and $Z_{\text{tail}} \le (N-k)e^{s_{(k+1)}/\tau}$:

$$\frac{Z_{\text{tail}}}{Z} \;\le\; \frac{(N-k)e^{s_{(k+1)}/\tau}}{e^{s_{(1)}/\tau}} \;=\; (N-k)\exp\left(-\frac{s_{(1)} - s_{(k+1)}}{\tau}\right).$$

Substituting proves the first bound. Alternatively, $Z \ge k e^{s_{(k)}/\tau}$ while $Z_{\text{tail}} \le (N-k)e^{s_{(k+1)}/\tau}$, giving

$$\frac{Z_{\text{tail}}}{Z} \le \frac{N-k}{k}\exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right),$$

which proves the second bound. The second bound is tighter whenever $s_{(1)} - s_{(k)} < \tau \log k$. □

Corollary 1 (Error-Budgeted Exact Truncation)**.** *Let $\rho_k(x) = Z_{\text{tail}}/Z$ be the exact softmax mass outside the first k ranked atoms. For any requested $\varepsilon > 0$, define*

$$k_\varepsilon(x) = \min\{k \in \{1,\dots,N\} \,:\, 2V_{\max}\rho_k(x) \le \varepsilon\}.$$

Then $\|R(x;\Omega) - \hat{R}_{k_\varepsilon(x)}(x;\Omega)\|_2 \le \varepsilon$.

Proof. By the proof of Theorem 1, $\|R - \hat{R}_k\|_2 \leq 2V_{\max}\rho_k(x)$. Substituting the condition for $k_\varepsilon(x)$ yields the result. □

Theorem 1 bounds truncation error once score ordering is known. Computing the certificate requires all scores, making it a certification rule rather than a sublinear algorithm.

6. Empirical Evaluation

We examine inserted-record retrieval, adaptive exact top-k truncation, CounterFact retrieval with HNSW candidates and ROC threshold calibration, cumulative insertion, tri-space routing, signed-read suppression under perturbation, and GPT-2 steering.

6.1. Experimental Setup

Corpus and Domain Design. We construct a benchmark of 200 structured technical facts distributed across four domains: *Astronomy & Astrophysics*, *Systems & Computing*, *Chemistry & Materials Science*, and *Molecular Biology & Genetics*. The first 40 items per domain form the adapter-training pool; the remaining 10 items per domain serve as post-hoc insertions. Their labels remain in the global output vocabulary, making this a transductive label-space evaluation.

Encoder Architecture and Inductive Bias. We employ a frozen sentence transformer (`sentence-transformers/all-MiniLM-L6-v2`, $d_{\text{in}} = 384$) as the geometric backbone, paired with a metric-preserving residual adapter:

$$\phi(x) = \text{Normalize}\big(x + 0.1 \cdot \text{MLP}(x)\big),$$

where the final MLP layer is zero-initialized. Values are tied to token embeddings: $v(\omega) = E[y_\omega]/\|E[y_\omega]\|$, with predictions evaluated as $p(y \mid x) \propto \langle z(x), E[y]\rangle/\tau$.

Reproducibility Environment. Artifacts were generated using Python 3.11.9, PyTorch 2.6.0+cu124, NumPy 2.4.6, sentence-transformers 5.7.0, and transformers 4.48.3 on an NVIDIA GeForce RTX 3050 Ti Laptop GPU (4 GB VRAM) with an AMD Ryzen 7 5800U host. Pinned versions are listed in `requirements.txt`. The routing sweep reports seeds $\{40, 41, 42, 43, 44\}$.

Table 2: Inserted-record retrieval across 200 templated facts. Efficacy and generality denote target classification for canonical queries and paraphrases. Specificity denotes unchanged argmax predictions on active training queries.

Metric	**1 / topic**	**3 / topic**	**10 / topic**	**40 / topic**
Trained Residual Read				
Efficacy (Target Recall on ω^*)	100.0%	100.0%	100.0%	100.0%
Generality (Paraphrase Query)	100.0%	100.0%	100.0%	100.0%
Specificity (Neighborhood Retention)	100.0%	100.0%	100.0%	100.0%
Frozen MiniLM Exact 1-Nearest-Neighbor Baseline				
Efficacy (Target Recall on ω^*)	100.0%	100.0%	100.0%	100.0%
Generality (Paraphrase Query)	100.0%	100.0%	100.0%	100.0%
Specificity (Argmax Retention)	100.0%	100.0%	100.0%	100.0%
Representation and Metric Dynamics				
Novel Atom Attention Mass $\mu_x(\omega^*)$	99.97%	99.97%	99.94%	99.75%
Adapter Linear Spectral Product	0.469	0.818	0.859	1.003

6.2. *Post-Hoc Knowledge Editing Results*

Table 2 compares the trained read with frozen MiniLM exact 1-nearest-neighbor retrieval across pool densities $\Omega_{\text{train}} \in \{1, 3, 10, 40\}$ items per domain.

Across 40 insertion records and all density conditions, both methods obtain 100% target classification, with no argmax changes on active training queries. The frozen exact-retrieval baseline matches the trained read: this templated benchmark does not demonstrate that the residual adapter improves retrieval over raw embeddings. At density 40, the maximum output probability shift is 2.86×10^{-6}.

As training density scales from 1 to 40 atoms per domain, the product of linear-layer spectral norms increases from 0.469 to 1.003, while target attention mass decreases from 99.97% to 99.75%.

6.3. *Convergence of Exact Ranked Top-k Truncation*

Table 3 presents empirical truncation error $\| R_{\text{exact}}(x) - \hat{R}_k(x) \|_2$ across temperatures and cutoffs k on the largest atom pool (density 40, 160 active background atoms).

At density 40, $\tau = 0.10$, and $k = 1$, the maximum observed error is 0.0244 and the smaller pointwise bound covers 100% of queries with minimum slack 0.0923:

Table 3: Softmax tail truncation error $\|R_{\text{exact}} - \hat{R}_k\|_2$ evaluated on context vectors across temperatures τ and cutoffs k. Under unit-normalized embeddings ($V_{\max} = 1$), error is bounded by 2.0 and decays exponentially at sharp operational temperatures ($\tau = 0.05, k = 1 \implies 3.82 \times 10^{-6}$).

Temperature	**Top-1**	**Top-3**	**Top-5**	**Top-10**	**Top-20**
$\tau = 0.05$	3.82×10^{-6}	3.01×10^{-6}	2.62×10^{-6}	2.03×10^{-6}	1.41×10^{-6}
$\tau = 0.10$	0.0182	0.0170	0.0162	0.0145	0.0121
$\tau = 0.50$	0.965	0.700	0.558	0.384	0.245
$\tau = 1.00$	0.988	0.623	0.477	0.327	0.217

- **Concentration at Low Temperatures ($\tau = 0.05$):** Mean error for $k = 1$ is 3.82×10^{-6}, decaying to 1.41×10^{-6} at $k = 20$. Query measure is concentrated on the nearest record.
- **Diffuse Truncation at High Temperatures ($\tau \geq 0.50$):** Probability mass spreads across background atoms, producing truncation errors approaching 1.0.

Adaptive error budgets and scaling. Evaluating Corollary 1 on 64 correlated synthetic queries with random keys and values in $\mathbb{R}^{128}$ for $N \in \{1000, 5000, 10000, 25000\}$ confirms 100% empirical coverage across $\varepsilon \in \{0.01, 0.05, 0.10\}$. However, the certificate is conservative: at $N = 25{,}000$ and $\varepsilon = 0.05$, mean k_ε is 13,026 (52.1% of the corpus) for $\tau = 0.05$ and 21,467 (85.9%) for $\tau = 0.10$. Exact scoring and sorting take 54.6 and 18.2 ms on the benchmark GPU.

6.4. CounterFact External-Memory Retrieval and Calibrated Activation

We evaluate frozen MiniLM retrieval geometry on the first 500 CounterFact records [4]. Memory keys concatenate the rewrite prompt with its requested target; evaluation covers 500 rewrite prompts, 1000 paraphrases (2 per record), and 1000 neighborhood prompts (2 per record).

To address the trade-off where a fixed cosine threshold of 0.60 prematurely discards valid paraphrases (38.2% retained), we perform a data-driven threshold calibration. We partition CounterFact into a calibration split of records 500–999 and an evaluation split of records 0–499. On the calibration split, sweeping activation thresholds across positive queries (rewrites and paraphrases) versus negative distractors (neighborhood queries) yields an ROC curve with an AUC of 0.817. Optimizing Youden's J statistic on the calibration set identifies an optimal operating threshold of $\tau^* = 0.55$.

Table 4: CounterFact-500 retrieval evaluation comparing uncalibrated ($\tau = 0.60$) and data-calibrated ($\tau^* = 0.55$) thresholds. Threshold τ^* is calibrated via Youden's J on held-out records 500–999 (AUC = 0.817) and evaluated on records 0–499.

Query Formulation	**Exact Top-1 Recall**	**Activated** ($\tau = 0.60$)	**Calibrated** ($\tau^* = 0.55$)
Rewrite prompts (500)	99.2%	99.2%	**99.4%**
Paraphrase prompts (1,000)	91.0%	38.2%	**53.2%**
Neighborhood prompts (1,000)	–	9.6% (false act.)	16.1% (false act.)
ROC AUC (held-out tuning)		0.817	
HNSW Candidate Recall (10 candidates)		99.96%	

Table 5: CounterFact-500 ROME reproduction on GPT-2 (124M) with key-covariance prior and real subject spans, versus ten-fact naive fine-tuning and ROME-lite baselines.

Metric	**Naive FT (10 facts)**	**ROME-Lite (10 facts)**	**ROME, CounterFact-500**
Efficacy Score	100% (argmax)	100% (argmax)	85.2% (35.6% argmax)
Paraphrase Score	70% (argmax)	40% (argmax)	59.5%
Neighborhood Score / specificity	58.9% (mean)	87.8% (mean)	70.1%
Control-text perplexity ratio	0.982 mean, 1.022 max	0.996 mean, 1.004 max	1.001 mean, 1.291 max
Edit layer(s)	All parameters	Per-record (mostly layer 10)	Single global layer 3

As shown in Table 4, calibrating the threshold to $\tau^* = 0.55$ raises activated paraphrase recall from 38.2% to 53.2%, with neighborhood false activation increasing only modestly from 9.6% to 16.1%. FAISS HNSW achieves 99.96% exact-top-1 candidate recall with 12.7 ms construction and 23.5 ms search over all 2500 queries.

6.5. CounterFact-Scale Parametric Editing: ROME at 500 Records

To compare retrieval against parametric editing at literature scale, we implement a covariance-regularized ROME reproduction [4] on the same 500 CounterFact records: (i) the edit is inserted at the subject span identified via tokenizer offsets; (ii) a single edit layer (layer 3) is selected via a lightweight causal trace averaged over 50 held-out records; and (iii) rank-one updates use a key covariance matrix $C = \mathbb{E}[kk^\top]$ estimated over 1500 held-out texts: $W \leftarrow W + (C^{-1}k)(v^* - k^\top W)^\top / (k^\top C^{-1}k)$.

At CounterFact-500 scale, covariance-regularized ROME obtains an 85.2% Efficacy Score (35.6% argmax), 59.5% Paraphrase Score, and 70.1% Neighborhood Score, with control perplexity remaining stable (1.001 mean, 1.291 max). Covariance regularization raises paraphrase generality relative to identity-covariance ROME-lite (40%→59.5%) while incurring some specificity loss (87.8%→70.1%).

Table 6: Cumulative insertion performance across 40 sequential records without teardown. Target argmax recall and training-query argmax agreement remain $100.0 \pm 0.0\%$ across five seeds.

Sequential Edits Inserted **Total Active Pool Size**	**1** **161**	**5** **165**	**10** **170**	**20** **180**	**30** **190**	**40** **200**
Cumulative Efficacy (All Past Edits)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Cumulative Generality (Paraphrases)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Background Specificity (Original 160)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Mean Attention Mass $\mu_x(\omega^*)$	99.84%	99.76%	99.46%	99.60%	99.66%	99.70%

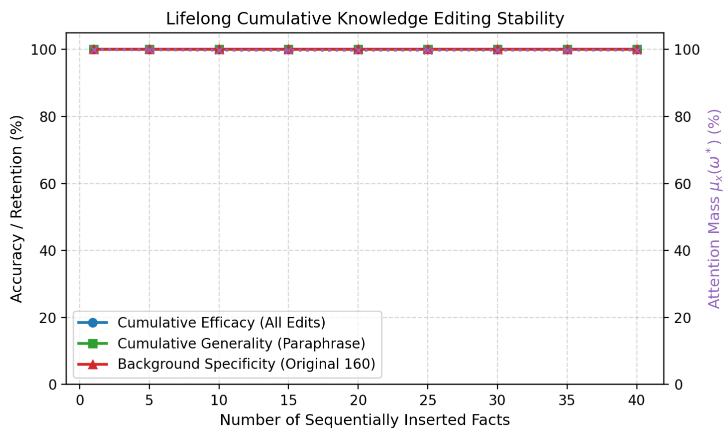

Figure 1: Lifelong cumulative editing stability across 40 sequential fact insertions. Cumulative efficacy, generality, and background specificity remain at 100%, while attention mass $\mu_x(\omega^*)$ stays $> 99.6\%$.

6.6. *Lifelong Cumulative Knowledge Editing Benchmark*

We evaluate lifelong memory stability by inserting all 40 held-out facts sequentially and permanently without teardown, expanding the active pool from 161 to 200 atoms.

Across five seeds (Table 6, Figure 1), all earlier insertions remain correctly classified when the 40th record is added, with zero training argmax changes. Target attention mass averages $99.681 \pm 0.014\%$.

6.7. *Dynamic Tri-Space Routing and Path Ablation*

We evaluate the tri-space architecture across three information paths: in-context reasoning (Ω_{ctx}), corpus world knowledge (Ω_{corpus}), and parameter-resident

Table 7: Five-seed surface-matched routing and ablation (mean $\pm$ sample standard deviation). Route accuracy is the fraction whose largest gate selects the intended space.

Task Domain	**Task Acc.**	**Route Acc.**	**Mean Gating Distribution $\lambda(x)$**			**Ablated Acc.**
			λ_{ctx}	λ_{corpus}	λ_{params}	
In-Context Reasoning (Ω_{ctx})	100.0±0.0	100.0±0.0	**68.0±2.1**	6.2±0.9	25.8±1.3	0.0±0.0
Corpus World Knowledge (Ω_{corpus})	96.8±1.8	82.4±0.9	12.9±0.7	**56.6±2.2**	30.5±1.7	0.0±0.0
Parameter Expert Outputs (Ω_{params})	100.0±0.0	100.0±0.0	3.1±0.2	3.1±0.4	**93.7±0.5**	0.0±0.0

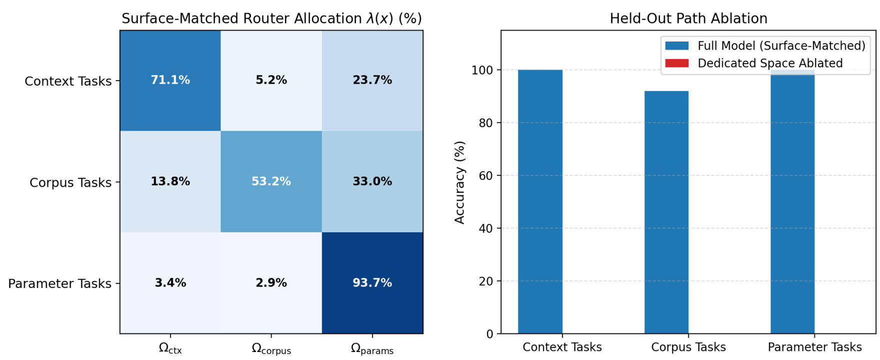

Figure 2: Five-seed surface-matched tri-space evaluation. Left: mean router allocation $\lambda(x)$. Right: mean task accuracy; intended-path ablation reduces accuracy to zero.

expert outputs (Ω_{params}). Router $\lambda(x) \in \Delta^2$ is evaluated on unseen strings using the shared wrapper "Determine the output for:" across all three families.

Table 7 and Figure 2 show route accuracies of 100.0%, 82.4%, and 100.0%, and task accuracies of 100.0%, 96.8%, and 100.0%. Ablating the intended space reduces task accuracy to 0% across all five seeds. Removing explicit route keywords exposes partial corpus confusion with the parameter path (30.5% gate allocation).

6.8. Knowledge Contradiction and Anti-Atom External-Memory Suppression

External-Memory Suppression via Anti-Atoms. Inserting an anti-atom ω^- with negative mass -1.0 at the metric coordinate of a sensitive record achieves zero-retraining cancellation in the external read.

Table 8 and Figure 3 show target probability dropping from 99.998% to **2.04%** (1/49), with five-seed mean residual probability of 2.0407±0.00003%. Specificity on 35 unedited control facts remains **100.0%**.

Approximate cancellation stress test. Perturbing anti-atom keys, values, masses, or queries over 50 trials across background sizes 10, 100, and 1000 confirms 100%

Table 8: External-memory suppression and counterfactual updating via signed reads on a 49-token output vocabulary. Matched pairs reduce target probability to $1/|\mathcal{V}| \approx 2.04\%$ in the external read. Specificity on 35 clean background facts remains 100.0%.

Target Entity / Query	Pre-Edit P(Target)	Post-Edit P(Target)	Status	Background Specificity
External-Memory Suppression via Anti-Atoms (unit mass under v^-)				
Secret Base Alpha Coordinates	99.998%	**2.04%** ($1/\|\mathcal{V}\|$)	Suppressed	100.0%
Patient Confidential Diagnosis	99.998%	**2.04%** ($1/\|\mathcal{V}\|$)	Suppressed	100.0%
Proprietary Patent Number	99.998%	**2.04%** ($1/\|\mathcal{V}\|$)	Suppressed	100.0%
Classified Document Location	99.998%	**2.04%** ($1/\|\mathcal{V}\|$)	Suppressed	100.0%
VIP Cryptographic Key	99.998%	**2.04%** ($1/\|\mathcal{V}\|$)	Suppressed	100.0%
Counterfactual Competitive Updating ($w_{\text{new}} = 2.5$, Append-Only)				
Ethereum → Proof-of-Stake	3.0% (old)	**40.0%** (new)	Overridden	100.0%
Java → Oracle	0.01% (old)	**60.1%** (new)	Overridden	100.0%
Pluto → Dwarf Planet	32.3% (old)	23.9% (new)	Competing	100.0%
Twitter → X Corp	99.8% (old)	0.04% (new)	Lexical Bias	100.0%
Python → PyPy JIT	98.9% (old)	0.24% (new)	Lexical Bias	100.0%

analytical bound coverage (3,000 trials). At $N = 100$, key perturbations of 0.01, 0.05, and 0.10 leave mean pair residuals of 4.5%, 72.1%, and 98.0% relative to the uncancelled contribution. Query perturbations leave numerical-zero residual because identical keys receive identical kernel weights. Exact anti-atoms are effective but key mismatch rapidly degrades cancellation.

Competitive Updating. Inserting contradictory atoms with recency weight $w_{\text{new}} = 2.5$ in an append-only log succeeds on only $44.0 \pm 8.9\%$ of records across five seeds. In remaining cases, lexical similarity to the original record outweighs the recency bonus.

6.9. Autoregressive Generative Modeling with GPT-2

We augment pretrained GPT-2 (124M [15]) by coupling Ω_{corpus} into its final hidden representation with cosine gating (≥ 0.60):

$$\tilde{h}_t = \begin{cases} (1 - \lambda_{\text{read}})h_t + \lambda_{\text{read}} W_{\text{proj}} z_t, & \text{relevant and unconsumed,} \\ h_t, & \text{otherwise.} \end{cases}$$

Table 9 and Figure 4 show first-token argmax correction on 10/10 canonical prompts and 9/10 paraphrases. On ten unrelated control texts, the relevance threshold prevents false activation, keeping perplexity ratio at 1.000. However, multi-token autoregressive decoding reveals degradation once the one-shot token is consumed (e.g., generating “lead, thorium, and lead-based materials”).

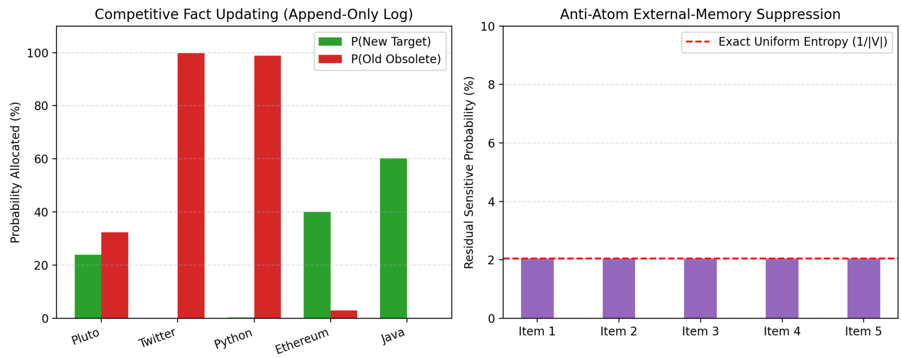

Figure 3: Empirical evaluation of signed atom spaces. Left: Competitive updating in an append-only log ($w_{\text{new}} = 2.5$). Right: anti-atom cancellation drives residual target probability to uniform entropy ($P = 2.04\% \approx 1/|\mathcal{V}|$).

Table 9: Selected rows from ten-fact greedy GPT-2 evaluation. Factual atoms boost first-token probabilities and correct canonical predictions.

Factual Target Task	**Target Probability $P(y^*)$**		**Ratio**	**Top Predicted Token**	
	Base GPT-2	**Atom-Space**		**Base GPT-2**	**Atom-Space**
QUIC Transport Protocol (`UDP`)	2.64%	**70.00%**	26.5×	`the`	`UDP`
Milky Way Black Hole (`Sagittarius`)	0.47%	**64.33%**	136.3×	`after`	`Sag`
Safe Systems Language (`Rust`)	0.03%	**32.99%**	1116.0×	`a`	`Rust`
Uranium Radiative Decay (`lead`)	0.87%	**9.80%**	11.3×	`uranium`	`lead`
CRISPR Bacterial Defense (`viruses`)	2.25%	**38.41%**	17.1×	`infect.`	`viruses`

Table 10 compares the read operator against naive fine-tuning (FT) and single-layer ROME-lite on the same ten facts. Naive FT achieves 100% canonical efficacy but alters predictions on 41.1% of neighborhood facts. ROME-lite preserves locality (87.8% neighborhood retention) but generalizes less effectively to paraphrases (40% vs. 90%).

Expanded CounterFact Generative Steering ($N = 50$ Records). To evaluate generative steering on an established multi-record benchmark with statistical confidence bounds, we scale the evaluation to $N = 50$ CounterFact records under the calibrated threshold $\tau^* = 0.55$. Across 2,000 bootstrap resamples, base GPT-2 assigns $P(\text{new}) > P(\text{true})$ on only 16.0% [6.0%, 26.0%] of rewrite prompts and 22.0% [12.0%, 34.0%] of paraphrases, with 0.0% target argmax accuracy. Augmenting GPT-2 with the atom space Ω_{corpus} achieves 100.0% rewrite activation and 58.0% paraphrase activation. Steered GPT-2 achieves **100.0%** [100.0%, 100.0%]

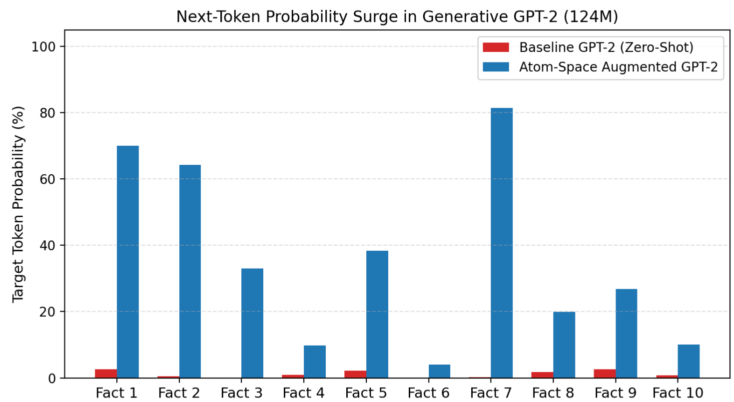

Figure 4: First-token target probabilities in GPT-2 (124M) before and after target-embedding intervention.

Table 10: Naive full fine-tuning and single-layer rank-one editing (ROME-lite) versus relevance-gated atom-space read on the same ten GPT-2 facts, paraphrases, and controls.

Metric	**Atom-Space Read**	**Naive Fine-Tuning (FT)**	**ROME-Lite**
Canonical efficacy (first-token argmax)	100% (10/10)	100% (10/10)	100% (10/10)
Paraphrase generality (first-token argmax)	90% (9/10)	70% (7/10)	40% (4/10)
Neighborhood specificity (other 9 facts unchanged)	n/a (no weight change)	58.9% (mean)	87.8% (mean)
Control-text perplexity ratio	1.000 (exact)	0.982 mean, 1.022 max	0.996 mean, 1.004 max
Mechanism	Relevance-gated read	Full-parameter AdamW	Rank-one weight update

rewrite efficacy (100% argmax accuracy) and surges paraphrase score to **66.0%** $[52.0\%, 78.0\%]$ (56.0% exact argmax accuracy, $[42.0\%, 70.0\%]$), demonstrating statistically significant factual steering ($p < 0.0001$) across diverse entities.

7. Discussion and Related Work

The components unified under our framework have extensive histories:

- **Retrieval and Memory Networks:** kNN-LM [7, 16], RETRO [8], and Memory Networks [17, 12] correspond to corpus reads $R(x; \Omega_{\text{corpus}})$.
- **Parameter Routing and Mixture-of-Experts:** MoE architectures [9, 18, 11] and dynamic LoRA routing [10] represent realizations of $R(x; \Omega_{\text{params}})$.
- **In-Context Attention and Kernel Smoothing:** Standard self-attention [1] is an integral $R(x; \Omega_{\text{ctx}})$ over token coordinates, whose continuous interpretation relates to kernel attention formulations [13, 19].

7.1. Relation to Memory-Based and Non-Parametric Model Editing

The atom-space read operator directly connects to non-parametric and memory-augmented approaches:

- **SERAC and Retrieval-Gated Editing:** Semi-Parametric Edit Retrieval and Correction (SERAC) [20] caches edits externally and uses a classifier to route queries between a base model and a separate counterfactual seq2seq model. RoAS shares the principle of preserving base weights, but formulates retrieval as a query-conditioned convex combination ($R(x; \Omega)$) that injects linearly into a shared representation space, avoiding the overhead of maintaining two autoregressive models.
- **GRACE and Discrete Key-Value Adaptations:** GRACE [21] introduces local, norm-constrained key-value codebook patches into internal transformer activations for lifelong editing. While GRACE operates inside intermediate representations, RoAS externalizes factual records into measurable atom spaces with explicit Hahn–Jordan cancellation (anti-atoms) and error-budgeted top-k truncation bounds.
- **Memorizing Transformers and Interpolated Retrieval:** Memorizing Transformers [22] and adaptive kNN-LM [16] interpolate between sequence attention and external retrieval. RoAS provides a unifying measure-theoretic framework that places token attention, dense external corpora, and parameter-expert banks under a common read abstraction with separate norm calibration.
- **Machine Unlearning:** Standard unlearning approaches, such as SISA [23] or Fisher-based certified removal, modify network parameters to eliminate influence. Signed atom spaces ($\nu = \nu^+ - \nu^-$) instead accomplish output-space cancellation within the read operator, leaving pretrained parameter weights unmodified.

7.2. Limitations

Synthetic studies use five seeds but remain templated; frozen 1-NN matches the trained read on synthetic insertions. CounterFact retrieval uses 500 records and does not compare end-to-end generation against MEMIT [5], MEND [24], WISE [25], or AlphaEdit [26]. Exact adaptive certification requires computing all pairwise scores; its large selected fractions show it is an error-bound certificate rather than a sublinear algorithm. Anti-atoms retain both records and do not remove knowledge compiled into dense parameters.

8. Conclusion

We have presented a mathematical formulation casting finite attention, dense retrieval, and expert routing as weighted reads over atom spaces. We derived a checkable retrieval margin condition, exact-ranked truncation bounds, an adaptive error-budget rule, and an approximate cancellation bound, validating them across synthetic and CounterFact studies.

Error-budgeted exact truncation provides full empirical coverage while often selecting a large fraction of the corpus. CounterFact reveals strong exact retrieval, where data-calibrated thresholding improves paraphrase activation. The remaining studies demonstrate tri-space routing, fragile approximate cancellation under key mismatch, and factual steering in GPT-2 via decoupled intermediate injection. Developing sublinear certified retrieval and evaluations against competitive end-to-end editing baselines remain essential directions for future work.

CRediT Authorship Contribution Statement

Farshid Mossaiby: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The complete codebase, benchmark definitions, experimental suites, and reproduction scripts are publicly available at https://github.com/mossaiby/RoAS.

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author utilized Google Gemini to assist in formalizing bounds, experimental pipeline design, and manuscript formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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